The 1+6+1 Partition of QED Selection Rules on S^3: Graph Topology, Angular Momentum, and the Dirac Spinor Phase Constraint

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Abstract

We classify a list of eight QED selection rules — five standard rules of perturbative QED and three specific to S^3 and the Camporesi–Higuchi construction — according to which structural ingredient is required to recover them on the GeoVac graph, within a vertex model (Wigner-3j coupling to scalar photon harmonics with an imposed parity rule; Paper 28). When QED is built on the scalar Fock graph with 1-cochain photons, only one of the eight rules survives—the Gaunt/Clebsch–Gordan sparsity built into the angular-momentum vertex. Building on the native Dirac graph with (n, κ, m_j) node labels scores three more (Δm_j conservation, spatial parity, and a graph analog of Furry's theorem) for a total of 4/8; these hold by the definition of that graph's edges. Promoting the photon from a scalar 1-cochain to a vector harmonic with explicit (q, m_q) angular labels and Wigner 3j vertex coupling recovers six angular-momentum rules at once (vertex parity, SO(4) channel count, the triangle inequality on n, Δm_j, spatial parity, Ward identity), with the vertex normalization contributing exactly $1/(4\pi)$ per loop (reading it as the S^2 Weyl exchange constant of the Hopf base is an Observation). This scores 7/8 for scalar electrons; the vertex-parity rule is the scalar-triangle zero of Paper 28's model, not a property of the physical vector vertex. The eighth rule is scored for Dirac electrons through a diagonal-vertex identity: for α_z against the scalar harmonic $Y_q^{m_q}$, the off-diagonal α-matrix and the i phase on the small component reduce $V(a, a, q, m_q) = V_{LS} + V_{SL}$ to an imaginary part that m_l conservation forces to vanish. This identity does not hold for a physical vector photon, whose $\alpha_\pm$ components give a nonzero diagonal (magnetic-moment) coupling, so it is not a recovery of Furry's theorem; in the production code the diagonal zero is imposed. The resulting tally, within the model, is 1+6+1:

1. Graph-topological (Gaunt/CG sparsity), surviving on any GeoVac discretization regardless of photon promotion.

6. Angular-momentum, recovered by promoting the photon to a vector harmonic, costing exactly $1/(4\pi)$ per loop.

1. Dirac-kinematic (the α_z diagonal-vertex identity), at no additional cost.

The tally is independent of the Paper 2 fine-structure-constant observation and is an observation [MEASURED, $n_{\max} = 2$] about this vertex model, not a recovery of QED's selection rules on S^3. All claims are verified by 103 tests in `tests/test_vector_qed.py` (including aggregate tallies of all four census columns) plus the 21-test symbolic Furry derivation in `tests/test_paper33_furry_derivation.py`.

1 Introduction

The census uses the eight selection rules of Table 1: five standard rules of perturbative QED and three specific to the S^3 construction.

#	Rule	Mechanism
1	Gaunt / CG sparsity	$3j$ symbols and triangle inequality on $\|l_1 - l_2\| \le l_\gamma \le l_1 + l_2$
2	Vertex parity / GS structural zero	$l_a + l_b + l_\gamma$ odd at the $\bar{\psi}\gamma^\mu\psi A_\mu$ vertex
3	SO(4) channel count	W-coefficients of the Camporesi–Higuchi spinor decomposition vanish on disallow
4	Δm_j conservation	Magnetic quantum number addition rule $m_a + m_\gamma = m_c$
5	Spatial parity (E1)	Photon is a vector field, parity -1; matrix element vanishes between same-parity
6	Ward identity	$k_\mu \mathcal{M}^\mu = 0$, gauge invariance
7	Charge conjugation (Furry)	Closed fermion loops with odd number of vertices vanish under C
8	Triangle inequality on SO(4)	$\|n_1 - n_2\| \le n_\gamma \le n_1 + n_2 - 2$

Table 1: The eight rules of this census. Rules 1, 4, 5, 6, 7 are standard rules of perturbative QED; rules 2, 3, 8 are specific to S^3 and the Camporesi–Higuchi construction, and rule 2 is the ground-state zero of Paper 28's vertex model (the physical vector vertex does not enforce it).

When QED is constructed on the GeoVac graph (Paper 28 [10] §graph_native_qed, Sec. 2 below), only Rule 1 survives by default. Recovering each remaining rule requires a specific structural promotion of the construction. This paper documents which promotion recovers which rule, with what calibration exchange constant.

Why this matters. In the spectral-triple framework (Paper 32 [13]), QED on S^3 is the spectral action of the GeoVac Dirac operator D_{GV}. The census asks which structural feature of the construction scores each rule in the vertex model. Cataloguing which rules require which features clarifies which selection rules are *universal* (Paper 22 [7] sense, depending only on $\mathcal{A}$) and which are *Coulomb-specific* (Paper 31 [12] sense, depending on D). The 1+6+1 partition is the empirical answer.

Scope. This is an observation paper about a vertex model. Five of the eight rules are standard; the GeoVac-specific content is which structural promotion scores which rule in the model, the per-loop $1/(4\pi)$, and the α_z diagonal-vertex identity, which is not Furry's theorem for a physical vector photon. No new physics, no fitting parameters; the result follows from explicit construction in `geovac/vector_qed.py` and verification in `tests/test_vector_qed.py` (103 tests, including aggregate tallies of all four census columns) and `tests/test_paper33_furry_derivation.py` (the symbolic Furry derivation, 21 tests).

2 The Graph-Native Baseline: 1/8

The graph-native QED construction of Paper 28 [10] §graph_native_qed builds all one-loop diagrams as finite matrix traces on the Fock graph. Photons are 1-cochains (functions on edges), photon propagator is the edge-Laplacian pseudo-inverse $G_\gamma = L_1^+$ where $L_1 = B^\top B$ is Paper 25's discrete Hodge-1 Laplacian, and the vertex is the bilinear $V(a, b, e) = \delta_{e,e_{ab}}$ enforcing that the two electron legs sit at the endpoints of the photon edge.

2.1 What survives

The only rule that survives this baseline is the Gaunt/CG sparsity of the angular-momentum vertex, which is built into the discrete edge set: the Fock graph has edges between (n, l, m_l) and (n', l', m_l') only if the standard ladder-operator selection rules are satisfied, which are exactly the

Gaunt/CG triangle conditions for the $\bar{\psi}\psi$ matrix element of an angular-momentum-1 vector field. This is Rule 1 in Table 1.

2.2 What does not

Rules 2–8 fail on the graph-native baseline. In particular:

Proposition 1 (Pendant-edge theorem)**.** *Let* $\Sigma_{\text{graph}}(\text{GS})$ *denote the one-loop electron self-energy on the scalar Fock graph evaluated at the ground state* $|1,0,0\rangle$*. Then for every* $n_{\max} \geq 2$*,*

$$\Sigma_{\text{graph}}(\text{GS}) = \frac{2(n_{\max}-1)}{n_{\max}} \xrightarrow{n_{\max}\to\infty} 2.$$

In particular $\Sigma_{\text{graph}}(\text{GS}) \neq 0$ *at every finite* $n_{\max}$ *and in the limit, in contrast to the ground-state zero of Paper 28's vertex model (Paper 28 [10] Theorem 4), which is itself not a property of the physical vector vertex.*

Mechanism. The ground state $|1,0,0\rangle$ is a pendant vertex (degree 1) of the Fock graph: its unique edge e_0 connects it to $|2,0,0\rangle$. The path-graph Laplacian inverse evaluated at this leaf edge gives $G_\gamma[e_0,e_0] = (n_{\max}-1)/n_{\max}$ exactly, and the vertex bilinear contributes a factor of 2; this is the only contribution to $\Sigma(\text{GS})$. Continuously verified in `tests/test_graph_qed_self_energy.py`: at $n_{\max}=2$ in exact rational arithmetic (the GS block is $[[1,1],[1,1]]$), and at $n_{\max}=3,4$ numerically to 10^{-10} (`TestPendantEdgeSweep`, added 2026-07-03 after the `graph_qed_photon` $n_{\max}\geq 4$ eigenvalue-sort crash fix; the exact-`sympy` pseudoinverse at $n_{\max}\geq 3$ exceeds any reasonable test budget, so those points are numeric). The closed form itself holds for all $n_{\max}$ by the path-graph Laplacian identity above. □

The model's mechanism (vertex parity: $l_a+l_b+l_\gamma$ odd forces zero at the GS because both electron legs are $l=0$) does not act on the scalar 1-cochain photon, which has no angular-momentum quantum number.

Conclusion. The graph-native baseline recovers 1/8 rules.

3 The Dirac Graph: 4/8

The native Dirac graph (CLAUDE.md §2, May 2026) replaces the scalar Fock node labels (n,l,m_l) with the relativistic Camporesi–Higuchi labels $(n_{\text{D}},\kappa,m_j)$, with two candidate adjacency rules: Rule A (κ-preserving) and Rule B (E1 dipole, $\Delta l=\pm1$). Rule B recovers spinor-intrinsic selection rules that scalar Fock nodes do not carry.

3.1 What is recovered

Three additional rules are recovered:

(**R4**) Δm_j conservation: built into the relativistic node label $m_j = m_l + m_s$, which is conserved at every electron–photon vertex with diagonal-in-m_j couplings.

(**R5**) Spatial parity (E1): the parity of a Dirac spinor is $(-1)^{l(\kappa)}$ where $l(\kappa) = -1-\kappa$ for $\kappa<0$ and $l(\kappa)=\kappa$ for $\kappa>0$. Rule B adjacency $\Delta l=\pm1$ enforces parity-changing transitions exactly as E1.

(**R7**) Charge conjugation (Furry): the off-diagonal Rule B vertex makes the identity-vertex tadpole vanish on the Dirac graph, which is a graph-level analog of Furry's theorem (the single-vertex closed loop is zero by edge-graph antisymmetry).

3.2 What is not

Four rules remain unrecovered on the Dirac graph alone:

(**R2**) Vertex parity / GS zero: requires the photon to carry an angular-momentum label l_γ, which the scalar 1-cochain does not.

(**R3**) SO(4) channel count: requires the photon W-coefficients of the Camporesi–Higuchi decomposition.

(**R6**) Ward identity: requires gauge invariance under the $U(1)$ phase of the photon, which the 1-cochain photon has only in temporal gauge and only approximately.

(**R8**) Triangle inequality: the graph adjacency is nearest-neighbor in n ($\Delta n = \pm 1$ only), but the continuum triangle allows $|n_1 - n_2| \leq n_\gamma \leq n_1 + n_2 - 2$ with $n_\gamma \geq 2$, exceeding the graph's reach.

Verdict. Dirac graph alone scores 4/8 rules. Of the four, R4, R5 and the graph analog of R7 hold by the definition of the Rule B edges ($\Delta l = \pm 1$, $|\Delta m_j| \leq 1$, bipartite by l-parity) and R1 by construction; they are construction-consistency checks, not computed recoveries. Three of the unrecovered rules (R2, R3, R6) are vector-photon-required; the fourth (R8) is a triangle-inequality range issue specific to the graph metric.

4 Vector-Photon Promotion: 7/8 (Scalar) and 8/8 (Dirac)

We now promote the photon from a scalar 1-cochain to a vector harmonic with explicit angular-momentum labels.

4.1 Construction

Definition 1 (Vector photon modes)**.** *For a maximum photon angular momentum $q_{\max} \geq 1$, the vector photon modes are the triples (q, m_q) with $1 \leq q \leq q_{\max}$, $-q \leq m_q \leq q$. The photon propagator is $G_\gamma(q) = 1/[q(q+2)]$, the eigenvalue of the vector Laplacian on S^3 at angular momentum q.*

Definition 2 (Wigner-3j vertex)**.** *For electron states $a = (n_a, l_a, m_a)$, $b = (n_b, l_b, m_b)$ and photon mode (q, m_q), the vertex coupling is*

$$V(a, b, q, m_q) = \langle l_a, m_a | l_b, m_b; q, m_q \rangle_{3j} \cdot \theta(l_a + l_b + q \ \textit{odd}) \cdot R(n_a, n_b, q), \tag{1}$$

where the 3j Clebsch–Gordan symbol enforces angular-momentum addition, $\theta(\cdot)$ is the parity step function (1 if true, 0 otherwise), and $R(n_a, n_b, q) = 1$ for the present analysis (we work at the spinor-amplitude level; radial integrals are absorbed into the 3j).

4.2 Selection rules recovered for scalar electrons

Theorem 1 (Scalar 7/8)**.** *With Definitions 1–2, the vector-photon QED on the Fock graph satisfies seven of the eight selection rules in Table 1, namely Rules 1, 2, 3, 4, 5, 6, 8. Rule 7 (Furry's theorem) fails: the vertex $V(a, a, q, m_q)$ is nonzero for $l_a \geq 1$ when q is odd, allowing nonzero closed-fermion-loop amplitudes with odd vertex count.*

Mechanism by rule. (R1) Built into the $3j$ symbol on $\langle l_a; l_b, q\rangle$.

(R2) The parity step $\theta(l_a + l_b + q \text{ odd})$ is exactly the vertex parity rule; at the GS $l_a = l_b = 0$, this forces q odd, which combined with the magnetic-quantum-number constraint $|m_q| \leq q$ and the Hermiticity of Σ produces $\Sigma(\text{GS}) = 0$ exactly.

(R3) The triangle inequality on the $3j$ symbol forces $|l_a - l_b| \leq q \leq l_a + l_b$; combined with the vector photon's full SO(4) harmonic content, this reproduces the continuum channel-count constraint. Census check (de-tautologized 2026-07-03): the per-(n_a, n_b) channel count realized in the vertex-tensor support matches the parity + triangle closed form for every pair (verified at $n_{\max} = 2, 3$).

(R4) The $3j$ symbol enforces $m_a + m_q = m_b$.

(R5) Built into the parity step.

(R6) At $n_{\max} = 2$ the Ward identity holds exactly (commutator ratio $\|[\Sigma, H_0]\|/\|\Sigma\| < 10^{-12}$); at $n_{\max} = 3$ it degrades (measured ratio ≈ 3.1) because cross-n couplings break the commutator metric on the truncated basis — the same degradation the Dirac census discloses (Theorem 2). The census is evaluated at $n_{\max} = 2$. (Corrected 2026-07-03: this item previously claimed a $\sim 10^{-3}$ residual at $n_{\max} = 3$, which the production diagnostic does not reproduce.)

(R7) FAILS: $V(a, a, q, m_q)$ for $a = (n, l \geq 1, m)$ and q odd contains terms proportional to $\langle l, m|l, m; q, 0\rangle_{3j} \neq 0$, allowing closed loops with odd vertex count.

(R8) Verified on the vertex-tensor support: every realized coupling satisfies $|n_a - n_b| \leq q \leq n_a + n_b - 2$ (census check `8_triangle_on_n`, added 2026-07-03). The upper bound follows from the parity step, the l-triangle, and $l \leq n - 1$; the lower bound is contingent and holds at the tested truncations $n_{\max} = 2, 3$.

□

4.3 The diagonal α_z vertex and Rule 7

Theorem 2 (Diagonal α_z vertex vanishes)**.** *Replacing the scalar electron states with Dirac spinors (large/small two-component decomposition with the standard i-phase on the small component, $\psi = (f\Omega_\kappa, ig\Omega_{-\kappa})$) and taking the vertex model's coupling, α_z against the scalar harmonic $Y_q^{m_q}$, the diagonal vertex bilinear satisfies $V(a, a, q, m_q) = 0$ identically. With it the model's census scores 8/8 at $n_{\max} = 2$ (the Ward-identity ratio degrades at $n_{\max} = 3$). The identity does not extend to a physical vector photon (Remark 2).*

Proof. **Step 1: $m_q = 0$ is forced.** For the diagonal matrix element $a = b$ we have $m_{j,a} = m_{j,b}$, and the Wigner-Eckart m-rule of Definition 2 gives $m_q = m_{j,b} - m_{j,a} = 0$. In the vertex model the photon is the scalar harmonic Y_q^0 and only α_z enters; we set $\alpha_z = \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix}$.

Step 2: bilinear evaluation. With the Dirac spinor $\psi = (f\Omega_\kappa, ig\Omega_{-\kappa})^\top$ and real radial functions f, g,

$$\begin{aligned}\alpha_z\psi &= (ig\sigma_z\Omega_{-\kappa}, f\sigma_z\Omega_\kappa)^\top,\\ \psi^\dagger\alpha_z\psi &= f^*\Omega_\kappa^\dagger(ig\sigma_z\Omega_{-\kappa}) + (-ig^*)\Omega_{-\kappa}^\dagger(f\sigma_z\Omega_\kappa)\\ &= ifg\big[\Omega_\kappa^\dagger\sigma_z\Omega_{-\kappa} - \Omega_{-\kappa}^\dagger\sigma_z\Omega_\kappa\big].\end{aligned}$$

Since σ_z is Hermitian, the bracketed quantity equals $M - M^*$ where $M = \Omega_\kappa^\dagger\sigma_z\Omega_{-\kappa}$, giving $\psi^\dagger\alpha_z\psi = -2fg\cdot\Im M$.

Step 3: $\Im M$ vanishes pointwise. Expand the spin-spherical harmonics in the standard Clebsch–Gordan basis, $\Omega_{j,l,m_j} = \sum_{m_s}\langle l, m_l; \tfrac12, m_s|j, m_j\rangle Y_l^{m_l}(\theta,\phi)\chi_{m_s}$ with $m_l = m_j - m_s$. The Pauli matrix $\sigma_z = \mathrm{diag}(+1,-1)$ is real-diagonal in the χ_{m_s} basis and conserves m_s. Therefore

$$M(\theta,\phi) = \sum_{m_s=\pm1/2}(\operatorname{sign} m_s)\langle l_\kappa, m_l; \tfrac12, m_s|j, m_j\rangle\langle l_{-\kappa}, m_l; \tfrac12, m_s|j, m_j\rangle Y_{l_\kappa}^{m_l\,*}Y_{l_{-\kappa}}^{m_l},$$

where $m_l = m_j - m_s$ is fixed by σ_z conservation and the Clebsch–Gordan coefficients are real. The product $Y_{l_\kappa}^{m_l\,*}(\theta,\phi)\,Y_{l_{-\kappa}}^{m_l}(\theta,\phi)$ has matched magnetic indices and the $e^{\pm im_l\phi}$ phases cancel exactly, leaving

$$Y_{l_\kappa}^{m_l\,*}Y_{l_{-\kappa}}^{m_l} = N_{l_\kappa,m_l}\,N_{l_{-\kappa},m_l}\,P_{l_\kappa}^{|m_l|}(\cos\theta)\,P_{l_{-\kappa}}^{|m_l|}(\cos\theta),$$

which is a real function of θ alone. Hence $M(\theta,\phi)$ is purely real, $\Im M \equiv 0$, and $\psi^\dagger\alpha_z\psi = 0$ pointwise on S^2.

Step 4: integral vanishes. Since the integrand is identically zero before integration, $V(a,a,q,m_q) = \int_{S^2}\psi^\dagger\alpha_z\psi\cdot Y_q^0(\theta,\phi)\,d\Omega = 0$ for every $q \geq 1$.

The proof is closed. Six concrete cases at $\kappa \in \{-1,+1,-2\}$ and $m_j \in \{\pm1/2, +3/2\}$ are verified symbolically in `tests/test_paper33_furry_derivation.py` (ported 2026-07-03 from the archived sprint driver): the bilinear $V_{LS}+V_{SL}$ simplifies to exactly zero pointwise on S^2 (Step 3), $\Im M \equiv 0$ with $M \not\equiv 0$ (the mechanism of Remark 1), and the projected integral $V(a,a,q,m_q) = 0$ sympy-exactly in every case. The derivation is independent of the production vertex code, whose diagonal short-circuit it backs. □

Remark 1 (The mechanism is m_l conservation, not Hermiticity)**.** *The cancellation in Step 3 comes from* two *structural inputs: (i) σ_z conserves m_l (because σ_z acts as a real-diagonal in the spin sector, leaving the spatial harmonic unchanged), and (ii) the matched magnetic index m_l on both sides of the bilinear forces the $e^{\pm im_l\phi}$ phases to cancel, producing a purely real θ-dependent function. Hermiticity of α_z alone does not suffice: it gives $V_{LS}+V_{SL} = 2i\,\Im(V_{LS})$, but to conclude that this vanishes we need the* additional *input that $\Im(V_{LS}) = 0$, which is the m_l-conservation argument above.*

Remark 2 (This is not Furry's theorem)**.** *Theorem 2 is a statement about the model vertex, not Furry's theorem. For a physical vector photon an $m_q = 0$ harmonic also has $m = \pm1$ spherical components that couple through $\alpha_\pm$, and the diagonal current $\psi^\dagger\boldsymbol{\alpha}\psi$ has nonzero azimuthal part: the diagonal magnetic-moment coupling is nonzero (for the ground state, $\kappa = -1$, $m_j = 1/2$, the projection $\int j_\phi\sin\theta\,d\Omega$ is $-4/3$ with the radial factor fg set to 1). Moreover, in a closed loop the intermediate vertices connect different states, so a diagonal identity removes only the tadpole; field-theoretic Furry, from the second-quantized C operation $\psi \to C\bar\psi^T$, cancels odd-photon loops to all orders. In `geovac/vector_qed.py` the diagonal zero is imposed by setting $V(a,a) = 0$; the symbolic test backs that choice for α_z only.*

5 The 1+6+1 Partition Theorem

Combining Sec. 2–4 gives the full classification.

Theorem 3 (1+6+1 tally in the vertex model — MEASURED at $n_{\max} = 2$)**.** *Within the vertex model, the eight rules of Table 1 group into three tiers by which ingredient of the GeoVac construction scores them.*

Tier I (graph topology, 1 rule). *Rule 1 (Gaunt/CG sparsity) is recovered by the graph-native baseline alone, on either the scalar Fock graph or the native Dirac graph, with or without vector-photon promotion. The rule is built into the discrete edge set as a triangle-inequality constraint on the $3j$ symbol.*

Tier II (angular momentum, 6 rules). *Rules 2, 3, 4, 5, 6, 8 (vertex parity,* SO(4) *channel count, Δm_j conservation, spatial parity, Ward identity, triangle inequality on n) are recovered* simultaneously *by promoting the photon from a scalar 1-cochain to a vector harmonic with explicit (q, m_q) labels and Wigner-$3j$ vertex coupling. This promotion contributes exactly one factor of $1/(4\pi)$ per loop (read below, Sec. 6, as the S^2 Weyl exchange constant of the Hopf base — an Observation). Rule 2 is the model's scalar-triangle zero.*

Tier III (Dirac kinematic, 1 rule). *Rule 7 is scored by the diagonal α_z identity (Theorem 2): the off-diagonal structure of $\boldsymbol{\alpha}$ reduces the diagonal bilinear to an imaginary part, and m_l conservation kills it (Remark 1). It holds in the model only; for a physical vector photon the diagonal coupling is nonzero (Remark 2).*

The total cost of recovering all eight rules from the graph-native baseline is:

$$cost(8/8) = \underbrace{0}_{\text{Tier I (free)}} + \underbrace{n_{\text{loops}} \cdot \tfrac{1}{4\pi}}_{\text{Tier II (per loop)}} + \underbrace{0}_{\text{Tier III (kinematic)}} = \frac{n_{\text{loops}}}{4\pi}.$$

Proof. By Theorems 1 and 2, together with Proposition 1 and Sec. 3. Verified numerically in `tests/test_vector_qed.py`: each of the four census totals 1/8, 4/8, 7/8, 8/8 is asserted as an aggregate per-rule tally (`TestCensusAggregates`, added 2026-07-03), and the Tier-III Furry mechanism is derived symbolically in `tests/test_paper33_furry_derivation.py`. □

6 The $1/(4\pi)$ Calibration Exchange Constant

The Tier-II promotion costs exactly $1/(4\pi)$ per loop. We read this as the S^2 Weyl exchange constant of the Hopf base (an Observation).

Proposition 2 ($1/(4\pi)$ identification)**.** *The vector-photon $l = 1$ self-energy on S^3 at $n_{\max} = 2$ equals exactly $1/(4\pi)$.*

Proof. Direct computation in `geovac/vector_qed.py`: the $l = 1$ self-energy is

$$\Sigma_{l=1}(n_{\max} = 2) = \sum_b \sum_{q,m_q} |V(a, b, q, m_q)|^2 \, G_e(b) \, G_\gamma(q) = \frac{1}{4\pi},$$

where at $n_{\max} = 2$ only $q = 1$ and the two $2p$ intermediate states b contribute, with $4\pi \sum_{m_q} |V|^2 = 9/2$ per b, $G_e = 1/3$ and $G_\gamma(1) = 1/3$, so the product is $2 \cdot \frac{9}{2} \cdot \frac{1}{9} \cdot \frac{1}{4\pi}$. Verified in test `test_scalar_l1_self_energy_is_c` (`tests/test_vector_qed.py`). □

Rule	Tier	Scalar Fock	Dirac graph	Vector photon (scalar)	Vector photon (Dirac)
1. Gaunt/CG	I	PASS	PASS	PASS	PASS
2. Vertex parity (GS zero)	II	FAIL	FAIL	PASS	PASS
3. SO(4) count	II	FAIL	FAIL	PASS	PASS
4. Δm_j	II	FAIL	PASS	PASS	PASS
5. Parity (E1)	II	FAIL	PASS	PASS	PASS
6. Ward	II	FAIL	FAIL	PASS	PASS
7. Furry	III	FAIL	PASS*	FAIL	PASS
8. Triangle on n	II	FAIL	FAIL	PASS	PASS
Total		1/8	4/8	7/8	8/8

Table 2: Selection-rule census across four GeoVac construction tiers. * On the Dirac graph alone, Furry holds because the off-diagonal Rule B vertex makes the identity-vertex tadpole vanish; this is a graph-level analog of Furry, structurally distinct from the single-particle Dirac spinor phase constraint of Theorem 2. Both give a zero tadpole; the α_z identity holds in the vertex model only (Remark 2).

6.1 Why $1/(4\pi)$

The Hopf base $S^2 = S^3/S^1$ is a round two-sphere (of radius $\frac{1}{2}$ for the unit S^3, area π); the solid-angle normalisation $1/(4\pi)$ is radius-independent (the unit-S^2 area is 4π). Wigner-3j matrix elements integrated over the full angular range produce a factor of $1/(4\pi)$ as the solid-angle normalization of the Weyl Hodge–star operator on the 1-form sector of S^2 [9, 1]. Phrased in the spectral-action vocabulary of [2] (the vocabulary, not a theorem of that reference), the per-loop factor $1/(4\pi)$ is read as the a_2 heat-kernel coefficient of the vector Laplacian on S^2, appearing as the Hopf-base contribution at each gauge-loop integration in the S^3 spectral expansion — GeoVac's own identification.

Observation 1 (Per-loop $1/(4\pi)$ as Paper 18 calibration)**.** *The factor $1/(4\pi)$ enters at each loop in the vector-photon construction. We read it as the S^2 Weyl exchange constant of the Hopf base, in the sense of Paper 18 [6] (a reading; the value is a vertex-normalization identity): a calibration exchange constant introduced by the projection of the spinor bundle onto the Hopf base. It is not derivable from the Coulomb potential alone and not derivable from the graph topology alone; it is the structural cost of scoring the angular-momentum selection rules in the vertex model on a discrete spectral construction.*

7 Connections to Other Papers

7.1 Paper 18 (exchange constant taxonomy)

The 1+6+1 partition of Theorem 3 maps cleanly onto Paper 18's three-tier exchange-constant taxonomy:

- Tier I (Gaunt/CG) is *intrinsic*: it lives in the discrete graph structure regardless of any continuous-limit choice.
- Tier II (angular momentum, 6 rules, costing $1/(4\pi)$ per loop) is *calibration*: the $1/(4\pi)$ is exactly Paper 18's characterization of a calibration exchange constant introduced by embedding the discrete graph in the continuum.

- Tier III (the α_z diagonal identity) is intrinsic to the model's spinor vertex; it is not a property of physical QED.

In Paper 18 language, within the vertex model six of the eight rules are calibration content, one is intrinsic to the graph, and one rests on the model's spinor vertex. This parallels the universal/Coulomb-specific partition of Paper 31 [12].

7.2 Paper 22 (angular sparsity theorem)

Paper 22's angular-sparsity theorem states that ERI density depends only on $l_{\max}$, not on the potential $V(r)$. In the language of the present paper this is the statement that Rule 1 (Gaunt/CG sparsity, Tier I) is universal: it does not depend on the choice of Dirac operator or even on the Coulomb structure. Theorem 3 extends this: in addition to the angular-sparsity rule that Paper 22 identified, six more rules can be recovered by promotion to vector photons, but at the cost of a calibration; only one selection rule is truly potential-independent.

7.3 Paper 28 (QED on S^3)

Paper 28's continuum spectral sums use the same vertex model (its Eq. for the vertex rule). The pendant-edge theorem (Proposition 1) is Paper 28's $\Sigma_{\text{graph}}(\text{GS}) \neq 0$ result; the vector-photon score on Rule 2 reproduces the model's $\Sigma(n_{\text{ext}} = 0) = 0$, which the physical vector vertex does not enforce.

7.4 Paper 32 (spectral triple synthesis)

In Paper 32's spectral-triple language, the present partition is the statement that:

- Tier I is enforced by the algebra $\mathcal{A}_{\text{GV}}$ of the GeoVac triple.
- Tier II is enforced by the rank-1 vector-bundle promotion of the photon to a 1-form section of the Hopf base, which is the inner fluctuation of D_{GV} in the abelian extension of Paper 25. The $1/(4\pi)$ is the heat-kernel a_2 coefficient of the vector Laplacian on the Hopf base.
- Whether Tier III has a counterpart in the real structure J_{GV} on $\mathcal{H}_{\text{GV}}$ is not shown; the model identity uses only the i phase on the small component and m_l conservation.

The present paper is therefore the spectral-action selection-rule audit of the GeoVac triple constructed in Paper 32.

8 Open Questions

Q1. Does the $1/(4\pi)$-per-loop calibration extend to two and three loops? At one loop the calibration is $1/(4\pi)$ per photon-vertex pair and the Tier-II cost is exactly $1/(4\pi)$. At two loops, naive counting gives $1/(4\pi)^2$ per closed photon sub-loop, but Paper 28's three-loop $O(N^3)$-factorization shows non-trivial cross-channel structure that may modify this. Whether the per-loop counting holds exactly or only at leading order is open.

Q2. Do all six Tier-II rules cost the *same* factor $1/(4\pi)$? Theorem 3 groups Rules 2–6, 8 together as Tier II because they are all recovered by the same structural promotion (vector photon). But it is possible that individual rules carry sub-leading corrections that distinguish them at higher loop order. A track-scale verification at two loops would either consolidate Tier II or split it.

Q3. What does the census give with the physical vertex? With the vector-photon coupling rules (which open $q = n_1 + n_2 + 1$) and the full $\boldsymbol{\alpha}$ coupling (which makes the diagonal vertex nonzero), rules 2 and 7 would need to be re-scored. Whether other QED selection rules—in higher-rank extensions, or in the rank-2 Breit sector flagged in Paper 25 §VII.B and Paper 32 Q5—also follow from analogous spinor phase constraints is an open structural question.

Q4. Where does the partition sit on the universal/Coulomb spectrum of Paper 31? Paper 31 partitions GeoVac results into universal (algebra-only) and Coulomb-specific (operator-dependent). The present 1+6+1 partition has Tier I universal, Tier II calibration (Coulomb-specific to S^3 via the Hopf base), and Tier III a property of the model's spinor vertex. The correspondence between the two partitions is clean for Tiers I and II; Tier III does not carry over to a physical vector photon.

9 Conclusion

We have tallied eight selection rules (five standard, three specific to S^3) by which structural ingredient of a GeoVac vertex model scores them: one is graph-topological (Tier I), six are scored together by vector-photon promotion with $1/(4\pi)$ per loop (Tier II, the ground-state rule being the model's scalar-triangle zero), and one by the α_z diagonal-vertex identity (Tier III), which is not Furry's theorem for a physical vector photon.

The partition is structural and verified by 103 tests in `tests/test_vector_qed.py` (each census total asserted as an aggregate per-rule tally) together with the symbolic Furry derivation in `tests/test_paper33_furry_derivation.py`. It does not depend on the Paper 2 observation about the fine-structure constant; it stands as an observation about the cost of recovering each QED selection rule on a discrete spectral construction.

Reading the $1/(4\pi)$ as the S^2 Weyl exchange constant places the model's angular-momentum sector in Paper 18's calibration tier (an Observation). The census with the physical vertex is open (Q3).

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